Problem 17.1.11

Sketch or graph the given region and its image under the given mapping.

$$|z| \leqq \frac{1}{2}, \quad -\frac{\pi}{8} < \operatorname{Arg} z < \frac{\pi}{8}, \quad w = z^2$$

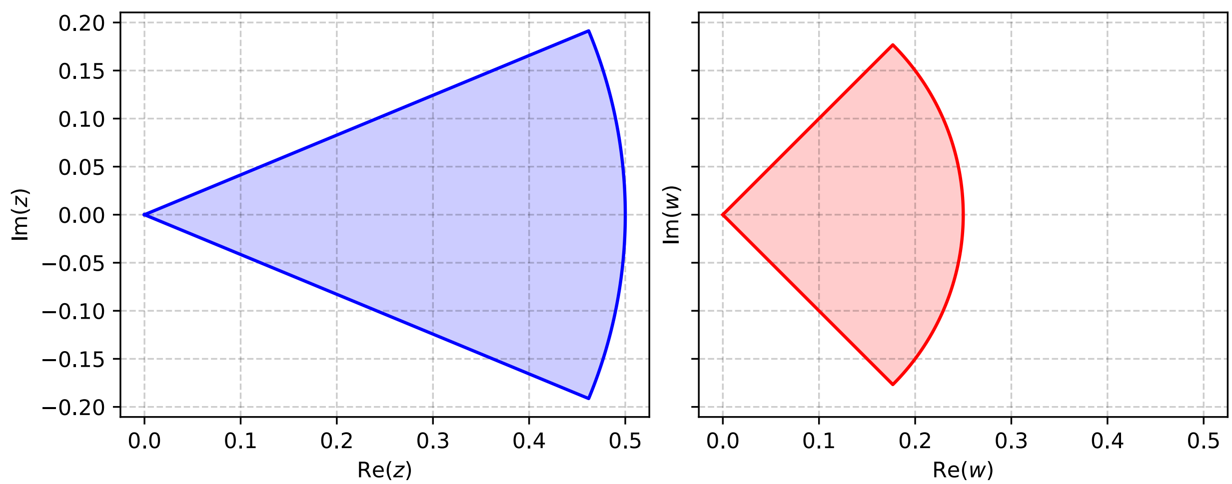
Solution

$$z = re^{i\theta}$$

$$w = Re^{i\phi}$$
$$z^2 = r^2 e^{2i\theta}$$

$$R = r^2 = \frac{1}{4}$$

$$\phi = 2\theta \implies -\frac{\pi}{4} < \operatorname{Arg} w < \frac{\pi}{4}$$



Problem 17.1.13

Sketch or graph the given region and its image under the given mapping.

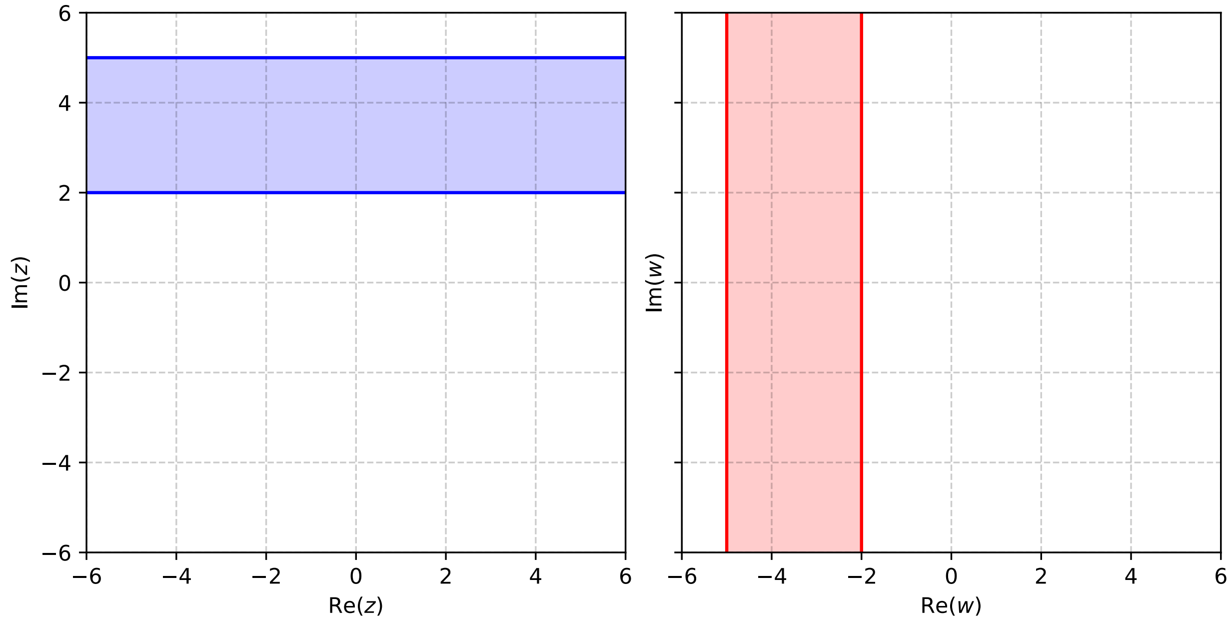
$$2 \leqq \operatorname{Im} z \leqq 5, \quad w = iz$$

Solution

The region is a strip bounded by $z_1 = 2i$ and $z_2 = 5i$, mapping to $w = iz$ gives another strip bounded by

$$w_1 = iz_1 = -2$$

$$w_2 = iz_2 = -5$$



Problem 17.1.15

Sketch or graph the given region and its image under the given mapping.

$$\left|z - \frac{1}{2}\right| \leqq \frac{1}{2}, \quad w = \frac{1}{z}$$

Solution

Problem 17.1.17

Sketch or graph the given region and its image under the given mapping.

$$\operatorname{Ln} 2 \leqq x \leqq \operatorname{Ln} 4, \quad w = e^z$$

Solution

Problem 17.1.19

Sketch or graph the given region and its image under the given mapping.

$$1 < |z| < 4, \quad \frac{\pi}{4} < \theta \leqq \frac{3\pi}{4}, \quad w = \operatorname{Ln} z$$

Solution

Problem 17.2.7

Find the inverse $z = z(w)$ Check by solving $z(w)$ for w.

$$w = \frac{i}{2z - 1}$$

Solution

Problem 17.2.13

Find the fixed points.

$$w = 16z^5$$

Solution

Problem 17.2.15

Find the fixed points.

$$w = \frac{iz + 4}{2z - 5i}$$

Solution

Problem 17.2.19

Find all LFTs with fixed point(s).

$$z = \pm i$$

Solution

Problem 17.3.9

Find the Linear Fractional Transformation (LFT) that maps the given three points onto the three given points in the respective order.

$$1,\ i,\ -1 \text{ onto } i,\ -1,\ -i$$

Solution

Problem 17.3.11

Find the LFT that maps the given three points onto the three given points in the respective order.

$$-1,\ 0,\ 1 \text{ onto } -i,\ -1,\ i$$

Solution

Problem 17.4.1

Find the image of $x = c = \text{const}$, $-\pi < y \leqq \pi$, under $w = e^z$.

Solution

Problem 17.4.3

Find and sketch the image of the given region under $w = e^z$.

$$-\frac{1}{2} \leqq x \leqq \frac{1}{2}, \quad -\pi \leqq y \leqq \pi$$

Solution

Problem 17.4.5

Find and sketch the image of the given region under $w = e^z$.

$$-\infty \leqq x \leqq \infty, \quad 0 \leqq y \leqq 2\pi$$

Solution

Problem 17.4.7

Find and sketch the image of the given region under $w = e^z$.

$$0 < x < 1, \quad 0 < y < \pi$$

Solution

Problem 17.4.11

Find and sketch or graph the image of the given region under $w = \sin z$.

$$0 < x < \frac{\pi}{2}, \quad 0 < y < 2$$

Solution

Problem 17.4.13

Find and sketch or graph the image of the given region under $w = \sin z$.

$$0 < x < 2\pi, \quad 1 < y < 3$$

Solution

Problem 17.4.19

Find the images of the lines mapping $x = c = \text{const}$ under the mapping $w = \cos z$.

Solution

Problem 17.4.21

Find and sketch or graph the image of the given region under the mapping $w = \cos z$ directly from **Problem 17.4.11**.

$$0 < x < \frac{\pi}{2}, \quad 0 < y < 2$$

Solution